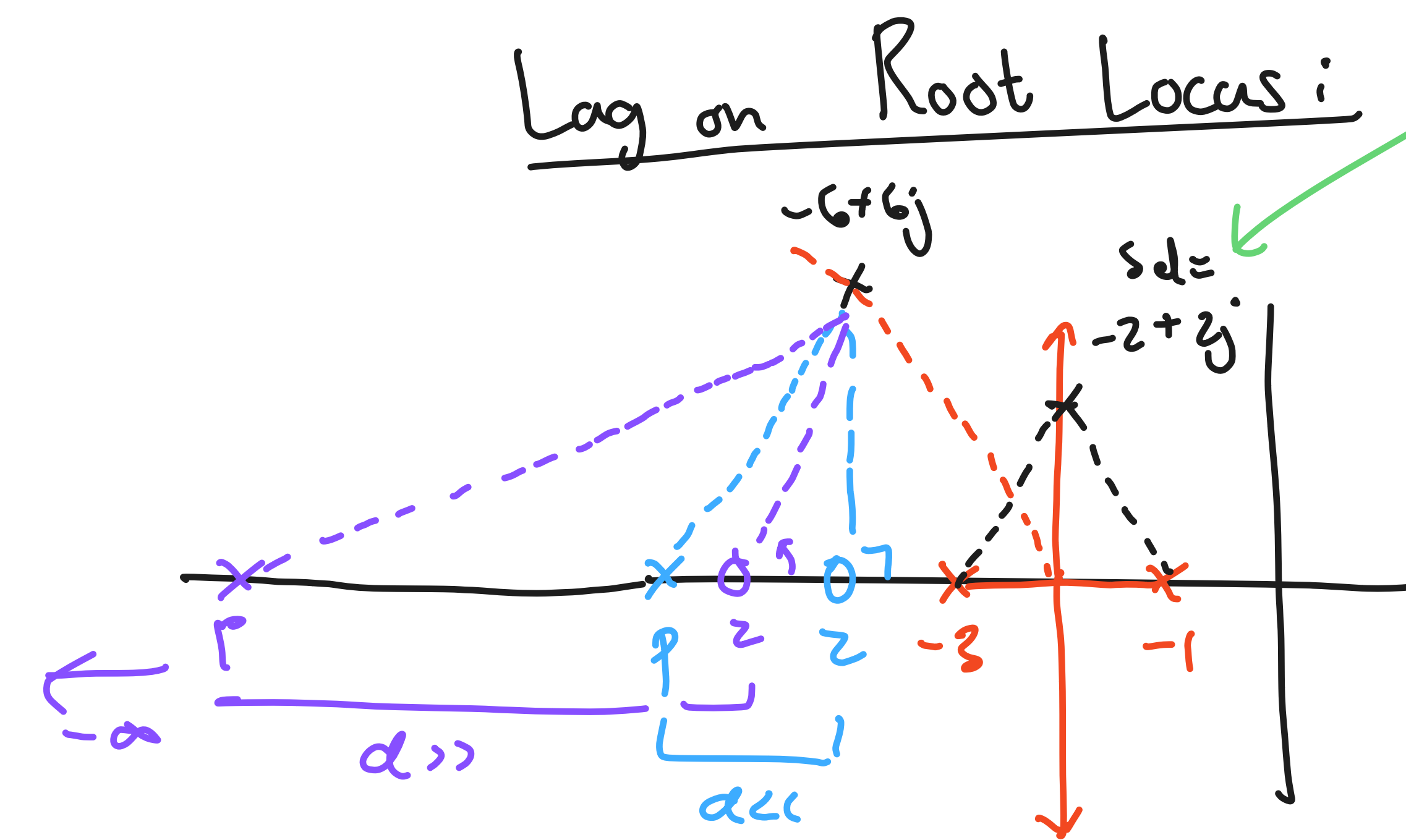




Lag Compensator $\approx$ integrator
 $\rightarrow$ for improving accuracy
 by increasing gain at low frequencies
 OR increasing gain at desired pole
 location on RL.

Lag Design:
 1) solve all your other
 problems with
 proportional / lead
 compensation

2) Work out how much
 extra gain you need
 and add a compensator
 $10 \rightarrow 50 \times$ closer to
 origin than your
 system's poles.



problem: we want gain
 here to be $10 \times$ higher.

what determines the gain
 at this location?

$$G(s) = \frac{1}{(s+1)(s+3)}$$

$$\therefore |G(s_d)| = \frac{1}{|s_d+1| \cdot |s_d+3|}$$

with compensator:

$$K = \frac{1}{|C(s_d)G(s_d)|} \quad \therefore \text{want more gain, want larger } \frac{1}{|C(s_d)|} \text{ i.e. smaller } |C(s_d)|$$

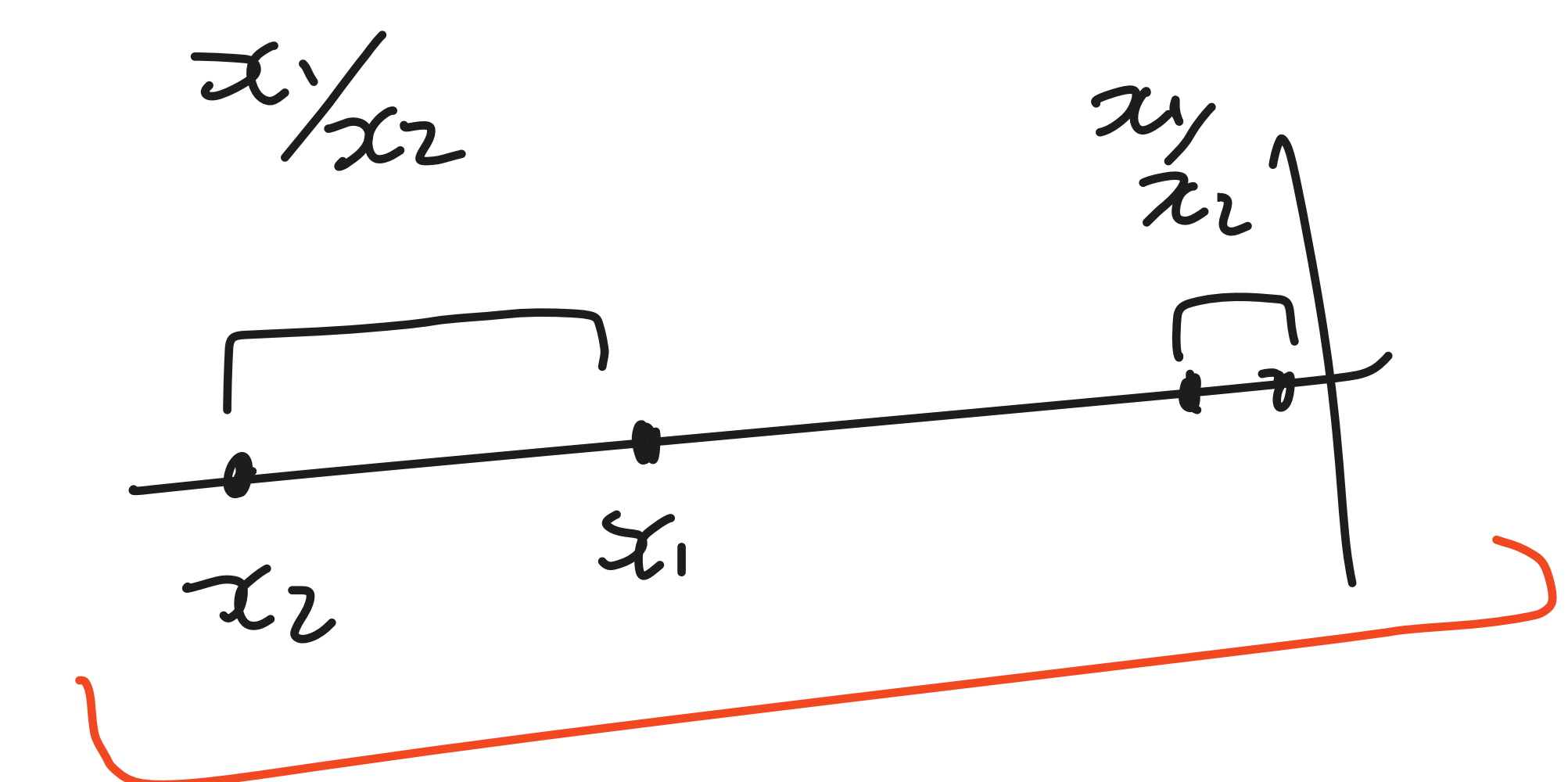
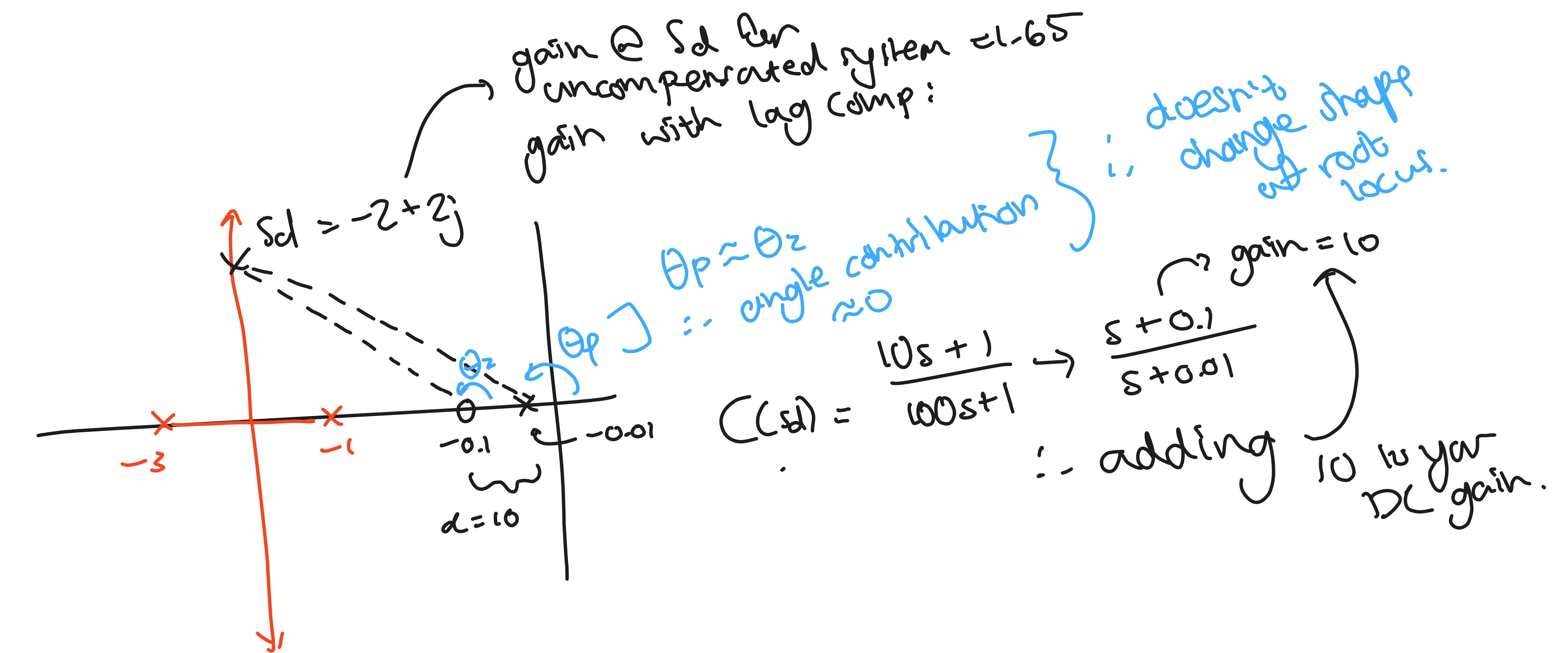
$$C(s) = \frac{1}{\alpha} \frac{s+1}{s+0.01} \rightarrow \infty \quad \therefore C(s_d) \text{ gets really small} \quad \therefore K \text{ gets really big.}$$

this gets bigger
this goes to ∞

$\therefore$ gain from the
 compensator
 at desired pole
 completely depends
 on pole-zero
 ratio.

$\therefore$ If you want to add
 extra gain, need to add
 a pole and zero with
 a large ratio ~~of~~
 between them.

$\rightarrow$ Ratio i.e. relationship
 of pole vs. zero is
 important, not
 absolute distance.



These distances are
 identical from a
 gain perspective.